PRAMANEX Bharat-VPop: A Governed Virtual Population Control Plane

DOI: https://doi.org/10.5281/zenodo.23238917

Bounded Evidence Ingestion, Simulation Traceability, and Multi-Gate Conjunction Release for Model-Informed Drug Development

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DOCUMENT CLASS
Control Plane Specification

EVIDENCE STATE
Controlled synthetic preprint

COMPILATION
Publication-ready architecture edition · Oct 2026

EXECUTIVE BRIEF

Drug development needs simulation populations whose origin, assumptions, subgroup coverage and privacy risk can be inspected. Clinical records remain fragmented by institution, country, coding system, sampling frame and access policy.

Bharat-VPop proposes a control plane that connects permitted evidence to context-specific virtual cohorts and model-informed questions. Its public method combines hierarchical representation, supported dependencies, constraints, subgroup uncertainty and downstream simulation adapters. The design can evolve toward privacy-preserving, multi-site evidence networks; those integrations and clinical outcomes are a roadmap, not an existing deployment.

The attached research source reports a controlled synthetic comparison, with 30 repetitions and three resulted methods. Its strongest synthetic fidelity and utility observations coexist with a material privacy proximity signal and a slightly worse constraint-violation rate than a Gaussian-copula comparator.

A platform cannot optimize realism alone; release must satisfy a strict conjunction of fidelity, utility, subgroup, constraint, privacy, provenance, and human disposition gates for a declared context of use.

No Indian population estimate, clinical validation, anonymity guarantee, EHR integration or regulatory acceptance follows from the synthetic results. The industrial ambition is a governed, reproducible simulation infrastructure whose evidence travels with every cohort and decision.

SOURCE-GROUNDED REFERENCE. Evidence boundaries, adverse findings, non-deployment statements, and scope limitations are preserved as first-class architecture constraints rather than relegated to footnotes.

1. THE INDUSTRIAL PROBLEM AND CONTEXT OF USE

An apparently realistic patient row is a weak unit of scientific evidence. Sampling, geography, care access, diagnostic definitions, calendar time, measurement and ancestry alter the population represented. A model can reproduce margins and still distort joint relationships, underrepresent a clinically relevant subgroup or disclose proximity to source members. Bharat-VPop treats a virtual population as a versioned research object with an explicit decision question and bounded interpretation.

Each run declares the research question, geography and population, age/sex/disease strata, eligibility rules, intervention and model version, simulated output, evidence cutoff, acceptance criteria and named reviewer.

BOUNDARY CONSTRAINTS

Diagnosis, individual treatment, replacement of clinical trials and automatic regulatory conclusions are excluded from the presently supported use. The name Bharat describes the initial scientific motivation; it does not establish representativeness for India as a whole.

2. EVIDENCE AND INTEROPERABILITY PLANE

A future production data engine would admit only records with documented legal authority, institutional permission and purpose. FHIR and OMOP are candidate interoperability mappings for electronic health records and registries. Their use would standardize representations, not prove population validity.

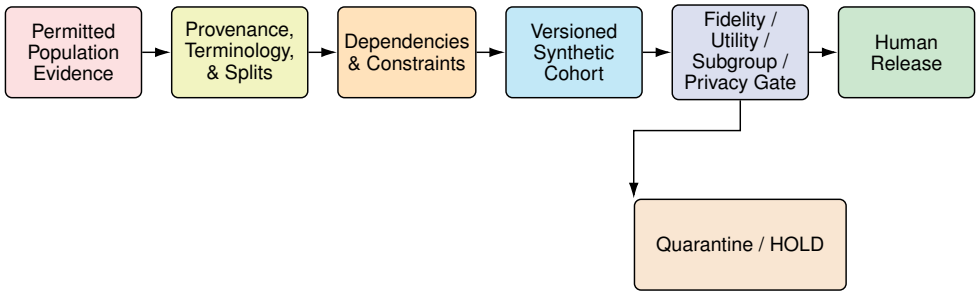
Cross-border intake would require jurisdiction-specific access, retention, transfer and ethics review. Multi-tenant boundaries must prevent raw record, feature, embedding and cohort leakage between organizations.

The proposed pipeline records original dataset identity, provenance, terminology mappings, inclusion/exclusion logic, observation period and transformations before modeling. Missingness, incompatible units, inconsistent disease definitions and unverified joins become quarantine cases. Data minimization and aggregate or federated analysis should be evaluated where direct pooling is impermissible.

PLANE CONTROL	BOUNDED ARCHITECTURAL RULE
FHIR / OMOP	Candidate mappings for EHR and registry interoperability; standardization does not prove population validity.
Cross-border intake	Jurisdiction-specific access, retention, transfer and ethics review precede use.
Multi-tenant isolation	Prevent leakage of raw records, features, embeddings and cohorts between organizations.
Data minimization	Evaluate aggregate or federated analysis where direct pooling is impermissible.

Current evidence boundary. No live FHIR, OMOP or registry connection is evidenced in the supplied study; these are architectural interfaces to build and test.

FIGURE 1. CONTEXT-BOUND COHORT CONSTRUCTION



PERMITTED EVIDENCE TO CONTROLLED DISPOSITION

Proposed architecture | failures enter HOLD / quarantine. The EHR, FHIR, OMOP and registry interfaces are target integrations. The reported experiment used generated synthetic records, not a live clinical network. Source rights and the context-of-use declaration precede any modeling.

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SOURCE_RIGHTS + CONTEXT_OF_USE → HARMONIZATION →
DEPENDENCY/CONSTRAINT MODEL →

VERSIONED_COHORT → CONJUNCTIVE_GATES → HUMAN_DISPOSITION
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The design makes two separations explicit: interoperability is not evidence validity, and synthetic appearance is not privacy safety. A versioned cohort is released only with its provenance, transformation history, declared scope, and gate evidence.

3. MODEL AND CONSTRAINT PLANE

Independent marginals preserve selected totals but not necessarily dependencies. Copulas model statistical association but can struggle under high-dimensional sparsity or misspecified distributions. Flexible deep generators introduce memorization and constraint risks. Mechanistic virtual populations need parameter provenance and model qualification.

Bharat-VPop's public design selects supported dependencies, distinguishes hard from soft constraints, represents subgroup uncertainty and exposes a downstream adapter boundary. The exact generator and constraint configuration were not included with the source PDF.

Hard / Soft / Unsupported

An accepted cohort must include a model and evidence manifest. Hard violations require rejection or repair under a documented rule, with the pre-repair output retained for evaluation. Soft constraints are reported with their uncertainty and trade-offs. Sparse or unsupported strata receive an explicit abstain state; synthetic sample size cannot manufacture empirical coverage.

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SUPPORTED_STRATUM = evidence sufficient   UNSUPPORTED_STRATUM =
                    explicit ABSTAIN

HARD_VIOLATION = reject OR documented repair + retained pre-repair
                 output
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A pharmacometric simulation adapter must carry its own qualified model assumptions and cannot inherit qualification from an upstream cohort fidelity score.

4. T-R-A-C-E SIMULATION LOOP

Track

Establishes permitted source, identity separation, time window, version and intended research purpose. Unknown rights or provenance block generation.

Resolve

Harmonizes units and terminology while preserving original values and conflicting definitions; unsupported joins enter quarantine.

Assign

Binds a population/model owner and qualified scientific, privacy and domain reviewers to the declared context.

Control

Applies deterministic identity separation, constraint, disclosure and access gates before any cohort release.

Evaluate

Measures fidelity, downstream utility, subgroup behavior, distribution shift, privacy and constraint failures with declared denominators.

SERVICE	PROPOSED RESPONSIBILITY
Ingestion	Issue immutable manifests and isolate ingestion failure from a true zero-result observation.
Normalization	Produce typed, terminology-aware artifacts while preserving original values and conflicts.
Validation	Compute separate scorecards for fidelity, utility, subgroup, shift, privacy and constraints.
Human Release	Approve a version-bound package; no automated model score grants release.

This is a proposed microservices decomposition, not evidence of an implemented high-throughput runtime. Queues and idempotent retry would isolate ingestion failure from a true zero-result observation. A failed critical gate quarantines the cohort and its derivatives, alerts the owner, records the cause and permits resubmission only after a new version and retest.

5. ZERO-TRUST PRIVACY AND GOVERNANCE MATRIX

Identity, location, data format and synthetic appearance each answer different questions. Authentication does not grant cohort access; a cross-border endpoint does not grant transfer permission; FHIR/OMOP mapping does not prove consent or validity; zero exact-row matches do not prove anonymity.

The reference design demands tenant-scoped keys and encryption, least-privilege service identities, purpose-scoped access, event-level audit, retention controls and independent review of disclosure risk. Specific key-management configurations and legal conformance remain deployment decisions.

INVARIANT	CONTROL STATEMENT
Tenant-scoped cryptography	Keys and encryption remain tenant-scoped; implementation details are deployment decisions.
Least privilege	Service identities and data access are restricted to minimum required purpose.
Purpose scope	Authentication alone does not grant cohort access or transfer permission.
Event audit	Material access and release actions are recorded at event level.
Retention	Retention controls are bound to context, rights and jurisdiction.
Disclosure review	Independent disclosure-risk review is a release-domain requirement.

Jurisdictional boundary

India's Digital Personal Data Protection framework, EU privacy and AI requirements, and relevant quality or computer software assurance concepts require applicability assessment by jurisdiction, data type and intended use. A design diagram cannot natively satisfy a law.

FDA's computer software assurance guidance is specifically scoped to medical-device production and quality management systems; it is an adjacent risk-based assurance reference, not a universal qualification of clinical simulation software. Cross-border evidence processing demands an organization-specific legal, privacy and ethics determination.

GOVERNANCE INVARIANT

Interoperability, authentication, zero exact-row matches, or synthetic appearance cannot be promoted into consent, legal transfer authority, empirical validity, or anonymity.

6. THE REPORTED SYNTHETIC EXPERIMENT & BENCHMARKS

The source describes 30 deterministic repetitions. Each uses a synthetic reference of 18,000 records, a biased source of 4,800 disjoint identities, a target of 6,000 and generated cohorts of 6,000 per method. A shifted synthetic test set has 8,000 records.

30 DETERMINISTIC REPETITIONS	**18,000 SYNTHETIC REFERENCE RECORDS**
4,800 BIASED SOURCE · DISJOINT IDENTITIES	**6,000 TARGET RECORDS**
6,000 GENERATED COHORT / METHOD	**8,000 SHIFTED SYNTHETIC TEST RECORDS**
3 METHODS WITH RESULT TABLES	**0 EXTERNAL POPULATION VALIDATIONS**

REPRODUCIBILITY LIMITATION

The source/target/holdout split and exact generator artifacts were not supplied. Numerical claims are author-reported and were not independently reproduced. Raw per-repetition outputs and analysis scripts were also not supplied; confidence intervals were not recalculated independently.

Three methods have result tables: independent marginals, Gaussian copula and the Bharat-VPop constrained method. The source also lists empirical resampling, Bayesian networks, CTGAN/TVAE, a constraint-free generator and module ablations in its wider protocol, but supplies no corresponding executed result tables. They are planned comparators, not observed evidence.

The report also describes a synthetic distribution-shift AUC of 0.936 and Brier of 0.094 for Bharat-VPop, versus 0.932 and 0.102 for copula. None of these values is an external population validation.

6.1 THE MASTER PERFORMANCE EVALUATION MATRIX

Metric	Independent Marginals	Gaussian Copula	Bharat-VPop Constrained
Standardized Wasserstein distance	0.119 (95% CI 0.115-0.124)	0.117 (95% CI 0.112-0.123)	0.031 (95% CI 0.028-0.033)
Correlation MAE	0.254 (0.253-0.255)	0.045 (0.043-0.047)	0.015 (0.013-0.016)
Subgroup diabetes-prevalence MAE	0.200 (0.196-0.203)	0.073 (0.070-0.076)	0.042 (0.039-0.044)
Downstream AUC	0.537 (0.457-0.617)	0.932 (0.931-0.933)	0.936 (0.934-0.937)
Brier score	0.218 (0.215-0.221)	0.105 (0.104-0.106)	0.093 (0.092-0.094)
Privacy proximity proxy AUC	0.543	0.539	0.825
Constraint violations	0.00000	0.00194	0.00237

Point estimates are shown without confidence intervals where none were supplied in the source. Lower is favorable for Wasserstein distance, MAE, Brier score and constraint-violation rate; the privacy-proximity AUC is an adverse-risk proxy and is not a utility target.

Reported paired effects versus Gaussian copula

WASSERSTEIN	Decrease 0.087; 95% CI 0.081-0.092
CORRELATION MAE	Decrease 0.031; 95% CI 0.029-0.032
SUBGROUP MAE	Decrease 0.031; 95% CI 0.028-0.035
DOWNSTREAM AUC	Increase 0.0034; 95% CI 0.0029-0.0039

These intervals were reported by the source and were not independently recalculated because raw per-repetition outputs and analysis scripts were not supplied.

6.2 PRIVACY PROXIMITY & ADVERSE SIGNAL ANALYSIS

RELEASE-BLOCKING ADVERSE SIGNAL

Bharat-VPop's reported proximity-based membership-risk proxy AUC is 0.825 versus 0.539 for Gaussian copula, despite no exact full-row matches. Median member distance-to-closest-record is 0.199 versus 0.450. These observations block generic anonymity or privacy-safety claims.

Privacy / constraint observation	Independent Marginals	Gaussian Copula	Bharat-VPop Constrained
Privacy proximity proxy AUC	0.543	0.539	0.825
Exact full-row match	0.000	0.000	0.000
Median member distance-to-closest-record	0.602	0.450	0.199
Constraint-violation rate	0.00000	0.00194	0.00237

The source's proxy does not equal a formal membership-inference attack. Its adverse direction is still material: generated records from Bharat-VPop lay closer to source members, and the proxy distinguished membership more readily. Formal membership- and attribute-inference attacks, differential privacy and a quantified privacy budget were not executed.

The shifted synthetic test reported Bharat-VPop AUC 0.936 (95% CI 0.935-0.937) and Brier 0.094 (0.093-0.095), versus copula AUC 0.932 and Brier 0.102. It was not a real Indian-population external validation.

Counter-result must travel with the gains.

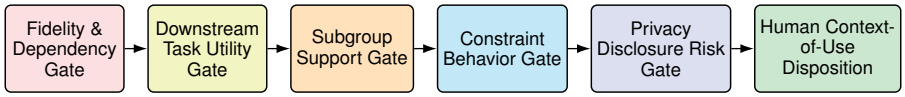
The constraint-violation result, 0.00237 versus copula 0.00194, must appear beside the fidelity and utility improvements because a headline metric can conceal a failed release domain.

7. RELEASE CONTRACT AND REPRODUCIBILITY

$$\text{RELEASE} = \text{FIDELITY} \wedge \text{UTILITY} \wedge \text{SUBGROUP_SUPPORT} \wedge \text{CONSTRAINT_BEHAVIOR} \wedge \text{PRIVACY_DISCLOSURE_RISK} \wedge \text{PROVENANCE} \wedge \text{HUMAN_DISPOSITION}$$

Release decisions must keep fidelity, utility, privacy, subgroup reliability and constraint behavior as separate domains. Thresholds must be prespecified for the intended use. A privacy failure cannot be averaged away by stronger fidelity; a subgroup gap cannot be hidden by global AUC. A reviewer receives model assumptions, failed cases, uncertainty, source rights and the exact cohort version. The output is a research artifact with limitations, not a clinical recommendation.

Figure 2. Conjunctive Release Gates



Proposed architecture | failures enter HOLD / quarantine. The gates are conjunctive rather than one weighted score. A privacy concern or unsupported subgroup blocks a release claim even when the synthetic utility metric improves.

GATE	DOMAIN	RELEASE RULE
01	Fidelity & dependency	Assess supported distributions and joint relationships. FAIL → HOLD
02	Downstream utility	Assess declared model-informed task behavior. FAIL → HOLD
03	Subgroup support	Preserve unsupported intersections as explicit abstain states. FAIL → HOLD
04	Constraint behavior	Expose hard and soft violations separately. FAIL → HOLD
05	Privacy disclosure risk	Do not substitute zero exact matches for privacy testing. FAIL → HOLD
06	Human disposition	Named reviewer binds release to the declared context of use. NO AUTOMATED RELEASE

Independent reproduction package

Independent reproduction requires the reference-population generator, source/target/holdout and shift-generation code, seeds, cohort or regeneration files, preprocessing, per-run outputs, analysis scripts, dependency lockfile and artifact hashes.

Before real-world use, a lawful, narrowly defined population must undergo representativeness and temporal/geographic external validation; formal privacy attacks and rare-record analysis; subgroup calibration and unsupported-intersection review; downstream model qualification; and independent replication. A finite benchmark cannot validate a universal national population model.

8. STRATEGIC TECHNOLOGY HORIZON

NEAR HORIZON	MIDDLE HORIZON	LONG HORIZON
Publish a reproducible benchmark and context-of-use registry, fix or explain adverse constraint/privacy cases, and demonstrate deterministic quarantine.	Under lawful partnerships, test controlled FHIR/OMOP mappings, permission-aware multi-site evidence, reviewer workflows and external population transportability.	Investigate a decentralized, continually re-evaluated simulation network with institution-controlled data, qualified domain models and version-bound release packages.

Real-time, autonomy and global reach are research and engineering targets contingent on data rights, scientific validation, privacy performance and accountable governance.

Durable founding thesis.

The cohort's value lies in the evidence, assumptions and limits that accompany a simulation, and in a release gate capable of refusing an attractive but unsafe result—even if the underlying generator is later replaced.

9. PRIVACY ATTACK AND SUBGROUP EVALUATION PROTOCOL

A future privacy evaluation should separate membership inference, attribute inference, record linkage, rare-stratum disclosure and nearest-neighbor proximity. Establish attacker access and auxiliary knowledge before running attacks. Calibrate results against a held-out reference and report uncertainty and subgroup-specific exposure.

Evaluate utility loss under any proposed privacy mechanism and state the privacy budget if differential privacy is used. A zero exact-match count is not a substitute for these tests.

For subgroup evaluation, predefine intersections, support thresholds, calibration measures and an unsupported state. Do not silently merge sparse strata to produce an attractive aggregate.

PROTOCOL DOMAIN	REQUIRED EVALUATION
Membership inference	Formal attack with explicit attacker access and auxiliary knowledge.
Attribute inference	Independent disclosure-risk test, not inferred from exact-match counts.
Record linkage	Assess linkage under realistic auxiliary information.
Rare strata	Report rare-stratum disclosure and subgroup-specific exposure.
Differential privacy	Report privacy budget if used and quantify associated utility loss.
Subgroup support	Predefine intersections, support thresholds, calibration and unsupported state.

10. OPERATIONAL FAILURE CASES

Reject a source whose purpose does not permit simulation; quarantine a unit conversion whose measurement basis is unknown; preserve a shifted cohort whose downstream task improves while an important subgroup worsens; and stop release when privacy proxy or hard-constraint gates fail.

A source update invalidates only dependent cohorts and simulations, but their prior outputs remain available for historical replay under the correct permissions. A reviewer correction becomes an annotated case for a new version, not an automatic generator change.

FAILURE CASE	REQUIRED DISPOSITION
Purpose does not permit simulation	Reject source before generation.
Unknown measurement basis	Quarantine unit conversion and preserve unresolved state.
Global task improves; subgroup worsens	Preserve evidence and block misleading aggregate release claims.
Privacy proxy or hard constraint fails	Stop release; cohort and derivatives enter HOLD / quarantine.
Source update	Invalidate dependent versions only; retain prior outputs for permissioned historical replay.
Reviewer correction	Create an annotated case for a new version; do not silently mutate the generator.

IMPLEMENTATION STATUS

These are proposed acceptance cases that require implementation and actual expected/observed test records.

11. STUDY INTEGRITY AND DECLARATIONS

The source reports generated synthetic records only, with no participants or patient records. Generative AI assisted manuscript structuring and language editing; the author retained scientific responsibility. The source says deterministic Python produced the outputs, but no code was supplied for verification.

The author is developing the concept, a potential competing interest; no external funding is reported. Future person-level health or genomic use requires lawful authority, purpose limitation, privacy and ethics review. These disclosures are preserved because they affect interpretation, not because they are administrative clutter.

CURRENT EVIDENCE STATE

The current evidence remains a controlled synthetic preprint. A production data engine, live cross-border clinical network, DPDP/EU legal conformance, validated pharmacometric model and autonomous development platform are proposed future states, not completed achievements.

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PRAMANEX Bharat-VPop is presented here as a governed reference architecture and synthetic benchmark interpretation. The document intentionally distinguishes observed benchmark evidence from future integrations, compliance determinations, external validation, and production deployment.

EVIDENCE STATE	RELEASE INVARIANT	DEPLOYMENT BOUNDARY
Controlled synthetic preprint; the reported study used generated synthetic records and no participants or patient records.	No automated model score grants release. Fidelity, utility, subgroup, constraint, privacy, provenance and human disposition remain separate gates.	Production data engines, live cross-border clinical networks, legal conformance, validated pharmacometric models and autonomous development remain proposed future states.

END OF PRAMANEX BHARAT-VPOP CONTROL PLANE SPECIFICATION